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Computer Science and Engineering Department

Computer Basic System Design

Concept Generation

Team Project

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1. Introduction

1.1 Purpose

The main purpose of this documents is to outline the detailed requirements specifications for the project.

1.2 Scope

This document specifies requirements for a device for managing the distance between means of transport and preventing traffic accidents with a primary focus on scooters.

1.3 Background

1.3.1 Theory behind the concept

Measuring distance and issuing alerts during emergency situations. Detecting and alerting is issued regardless of line of sight, ensuring safety scenarios where potential hazards may not be visually observable. This capability extends to alerting users of approaching vehicles, even when obscured from direct view, such as around corners. Detecting is covering multiple directions including left, right, front, rear, and corners.

1.3.2 Limitations of current designs or technology

Integrating distance detection technologies with existing vehicle systems or infrastructure can pose challenges, particularly in retrofitting older vehicles or adapting to diverse transportation environments.

1.3.3 Similarities and differences between our concept and existing technologies

The similarity between our design and current technologies lies in their shared capability of measuring distance and providing timely alerts to the driver in the event of approaching vehicles. However, a notable differentiation is the portability of our design, rendering it adaptable to a wide array of transportation environments. Additionally, our design is engineered to facilitate alerting across multiple directions, enhancing its versatility and effectiveness in diverse operational contexts.

2. Requirements Specification

2.1 Marketing Requirements

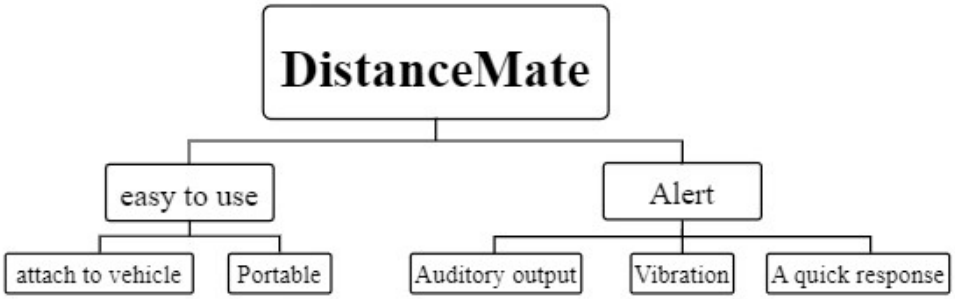
1. The device should have excellent alert quality.
2. The device should have recognizable vibration.
3. The device should be easy to use.
4. The device should be portable.
5. The device should be accessible to driver of all transportation modes, including scooters, bicycles, and motorcycles.
6. The device should have fast response time.
7. The device should have advanced technology.
8. The device should have flexible mounting.

2.2 Engineering Requirements

1. The device's respond time should be up to 5 seconds.
2. The device should identify objects in approximately 20 meters to front of it.
3. The device should identify objects to the side of it.
4. The device must have strong structural durability, assuring it to operate in various incidents.
5. The device shall have auditory output for emergency alert.
6. The device shall have vibration.

7. The device must recognize approximately 90% of approaching objects.
8. The device's sensor must not be giving false alerts.
9. The device shall measure the return time of ultrasound or laser by converting it into digital form.
10. The device must be able to distinguish between moving and non-moving objects.
11. The device must have structural protection.
12. The device's screen size should not exceed 4 inches.
13. The device must have 95% reliability for two years.

2.3 Objective Tree



2.4 Requirements Specification Table

Marketing Requirements	Engineering Requirements	Justification
6	1. The device's respond time should be up to 1 seconds.	This can be achieved by sensor technology. The sensors that can be used are as follows: EG laser, IR LED, Ultrasonic waves, etc.
1	2. The device shall have auditory output for emergency alert.	This can be achieved by sensor system, alert generation, or signal processing.
1, 2	3. The device shall have vibration.	This can be achieved by using vibration technology. Piezoelectric crystal that has a seismic mass can be used. The crystal is subjected to stress, an electric signal is generated.
3,4	4. The device should identify objects in approximately 20 meters to front of it. The device should identify objects to the side of it.	This can be achieved by distance sensor. Such as, Ultrasonic Sensor, Infrared Distance Sensors, Laser Distance Sensors: LIDAR, LED Time-Of-Flight Distance Sensors.
1, 7	5. The device must have strong structural durability, assuring it to operate in various incidents.	Outside protection: Weatherproofing (IP Ratings, IP67 or IP68), Anti-Glare and Anti-Reflective Coatings, Vibration and Shock

		Resistance, EMI Shielding. Inside protection: Heat Dissipation, Firmware and Software Robustness, Redundancy and Fail-Safe Mechanisms, Temperature Regulation.
6, 7	6. The device must detect approximately 90% of approaching objects.	Detection can be achieved by AI Integration, Data Integration, Radar Sensors, Data Fusion Module, etc.
1, 6	7. The device's sensor must not be giving false alerts.	This can be achieved by integrating high-quality sensors (miniaturized radar sensor), and advanced signal processing (noise and clutter reduction, sensor fusion), machine learning and AI.
7	8. The device shall measure the return time of ultrasound or laser by converting it into digital form.	Integration of microcontroller, or processor for processing data and running algorithms, and Analog-to-Digital Converter (ADC).
6, 7	9. The device must be able to distinguish between moving and non-moving objects.	This can be achieved by combining data, real-time processing AI, and motion analysis

		algorithms such as velocity calculation, and trajectory tracking.
3, 4, 8	10. The device's screen size should not exceed 4 inches.	Limiting the screen size to 4 inches, assures the need for a compact, portable, and power-efficient device that focuses on essential safety functions.
4, 5	11. The device must be able to attach to any vehicle.	By implementing above requirements, device will be able to use in any vehicle. To make the device to be able to attach on vehicle, following can be used: adjustable clamps, tool-free attachment, secure locking system, and multiple mounting options.
3, 5	12. The device must have 95% reliability for two years.	This can be achieved by high quality component selection (durable materials), robusting environmental protection, comprehensive testing and maitenance.

Marketing Requirements

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2.5 The Engineering-Marketing Tradeoff Matrix

		Identification Accuracy	Quick Response	Size	Internal structure
		+	-	-	+
Alert Quality	+	↑↑	↑		
Easy to Use	+		↑	↑↑	
Portable	+			↑↑	↓
Response Time	-	↑↑	↑↑		

2.6 Self-Assessment Checklist for the Requirements Specification

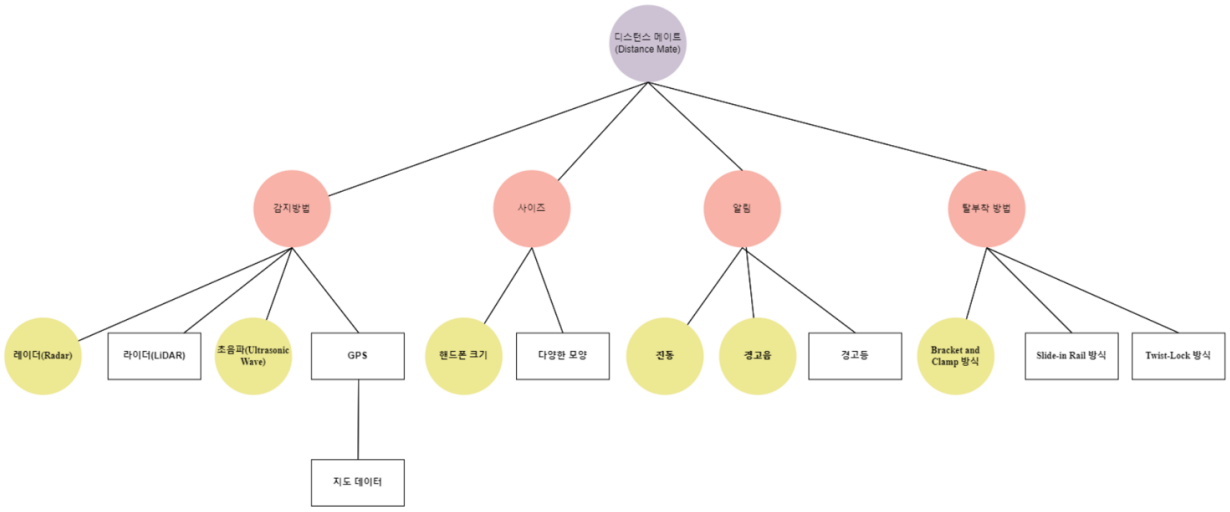
Engineering Requirements	Score
Each engineering requirement is abstract.	5
Each engineering requirement is verifiable.	3
Each engineering requirement is unambiguous and written as a concise statement.	4
Each engineering requirement can be traced to a user need.	5
Each engineering requirement is realistic and has justification provided.	5
Standarts and constraints applicable to the project have been identified and include.	4
The Requirements Specification	
The requirements are normalized, with minimal redundancy and overlap.	4
The engineering requirements are organized by similarity.	4
The requirements are complete, adressing all needs.	4
The requirements are bounded. (not overspecified).	5
The requirements have been validated and agreed upon by all stakeholders.	X

3. Concept Generation and Evaluation

3.1 디스턴스 메이트 (Distance Mate)의 컨셉 테이블

감지방법	사이즈	알림	탈부착 방법
레이더(Radar)	핸드폰 크기	진동	Bracket and Clamp 방식
라이더(LiDAR)	다양한 모양	경고음	Slide-in Rail 방식
초음파(Ultrasonic Wave)		경고등	Twist-Lock 방식
GPS (지도 데이터)			

3.2 디스턴스 메이트(Distance Mate)의 컨셉 팬



3.3 AHP를 위한 결정 매트릭스

	편리성	비용	안정성	정확성	weight
편리성	1	2	1/2	1/4	0.155
비용	1/2	1	1/3	1/4	0.086
안정성	2	3	1	1/3	0.262
정확성	4	4	3	1	0.497

		레이더	핸드폰크기	경고음	Bracket and Clamp 방식
편리성	0.155	0.21	0.10	0.29	0.40
비용	0.086	0.14	0.18	0.33	0.35
안정성	0.262	0.32	0.09	0.28	0.31
정확성	0.497	0.43	0.12	0.35	0.10
점수		0.342	0.114	0.321	0.223

3.4 강점, 약점 분석

method	strength	weakness
레이더(Radar)	장거리 감지 가능 다양한 날씨 조건에서 작동 가능	해상도가 낮을 수 있음 추가적인 하드웨어 필요
라이더(LiDAR)	높은 해상도 정확한 감지	가격이 비쌈 날씨 영향 받음, 설치 복잡
초음파(Ultrasonic Wave)	저속 주행시 정확한 감지 저렴한 비용	짧은 감지 거리 고속 주행 시 부적합
GPS (지도 데이터)	정확한 위치 정보 경로 기반 예측	지도 데이터의 정확도 업데이트를 해야하는 불편함
Bracket and Clamp 방식	견고한 브라켓과 클램프로 특정부위에 단단히 고정 가능 쉽게 탈부착 가능	초기 설치 시 브라켓을 고정하는 과정이 필요할 수 있음
Slide-in Rail 방식	단단히 고정가능 쉽게 탈부착 가능	레일을 설치해야함
Twist-Lock 방식	단다히 고정가능 쉽게 탈부착 가능	Twist-Lock 부품을 설치해야함